

● EASY

● ADVANCED MATH

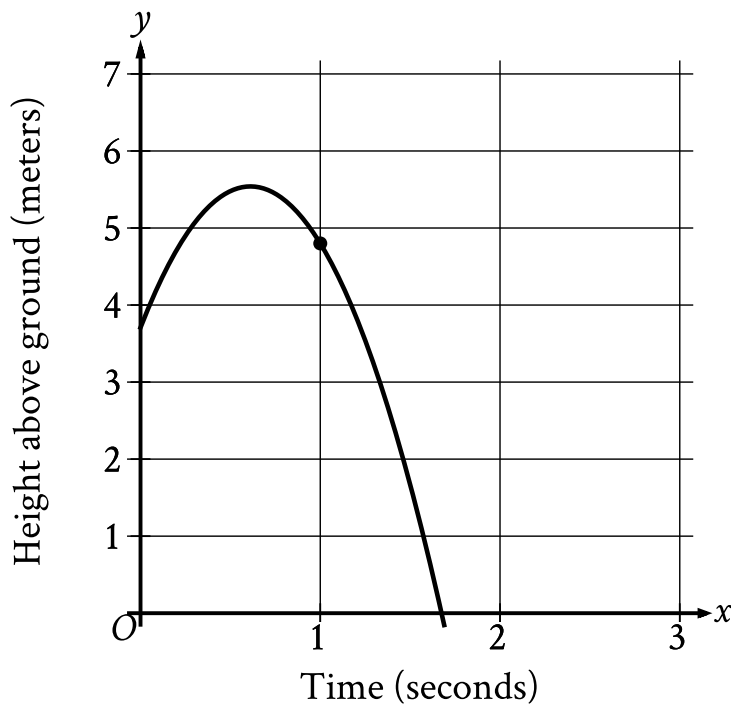
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# Nonlinear functions

10 questions · ~15 min

03



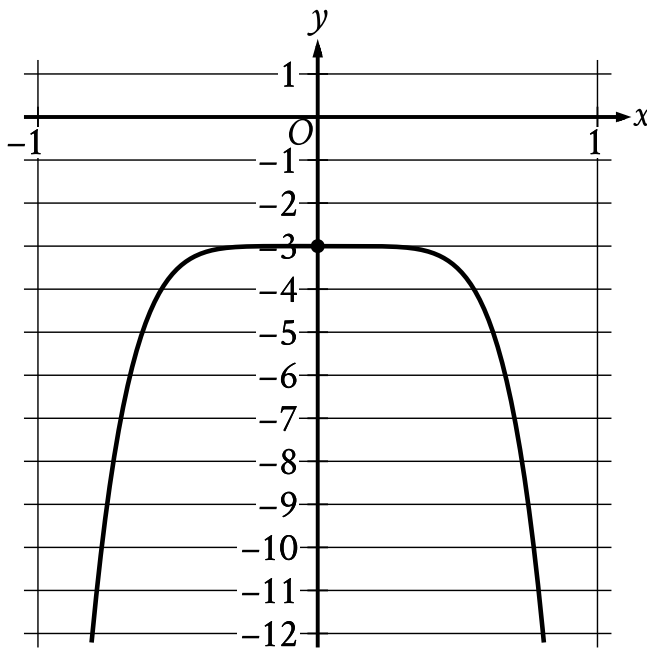


- All values are approximate.
- The parabola opens downward.
- The vertex of the parabola is at (0.6 comma 5.5).
- The parabola passes through the following points:
  - (0.0 comma 3.8)
  - (0.6 comma 5.5)
  - (1.0 comma 4.8)
  - (1.7 comma 0.0)

The graph shows the height above ground, in meters, of a ball  $x$  seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point 1.0, 4.8 in this context?

- A. 1.0 second after being launched, the ball's height above ground is 4.8 meters.
- B. 4.8 seconds after being launched, the ball's height above ground is 1.0 meter.

- C. The ball was launched from an initial height of 1.0 meter with an initial velocity of 4.8 meters per second.
- D. The ball was launched from an initial height of 4.8 meters with an initial velocity of 1.0 meter per second.



- In quadrant 3, the curve rises to its maximum at point (0 comma negative 3).
- In quadrant 4, the curve then falls.

The graph of the polynomial function  $f$ , where  $y = f(x)$ , is shown. The  $y$ -intercept of the graph is 0,  $y$ . What is the value of  $y$ ?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
| 1                    | 1                    | 1                    | 1                    |
| 2                    | 2                    | 2                    | 2                    |
| 3                    | 3                    | 3                    | 3                    |
| 4                    | 4                    | 4                    | 4                    |
| 5                    | 5                    | 5                    | 5                    |
| 6                    | 6                    | 6                    | 6                    |

|   |   |   |   |
|---|---|---|---|
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q3**

ADVANCED MATH

If  $f(x) = \frac{x^2 - 6x + 3}{x - 1}$ , what is  $f(-1)$ ?

- A. -5
- B. -2
- C. 2
- D. 5

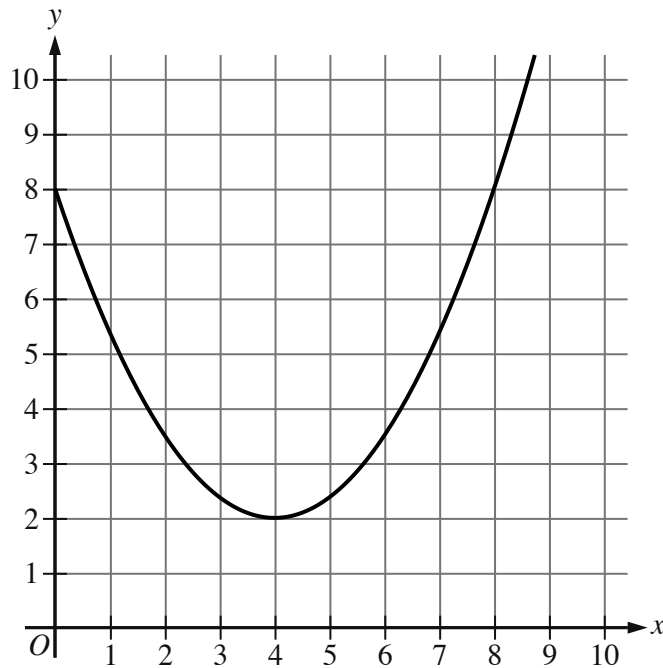
An investment account was opened with an initial value of \$ 890. The value of the account doubled every 10 years. Which equation represents the value of the account  $Mt$ , in dollars,  $t$  years after the account was opened?

A.  $Mt = 890 \frac{1}{2}^{\frac{t}{10}}$

B.  $Mt = 890 \frac{1}{10}^{\frac{t}{2}}$

C.  $Mt = 890 2^{\frac{t}{10}}$

D.  $Mt = 890 10^{\frac{t}{2}}$



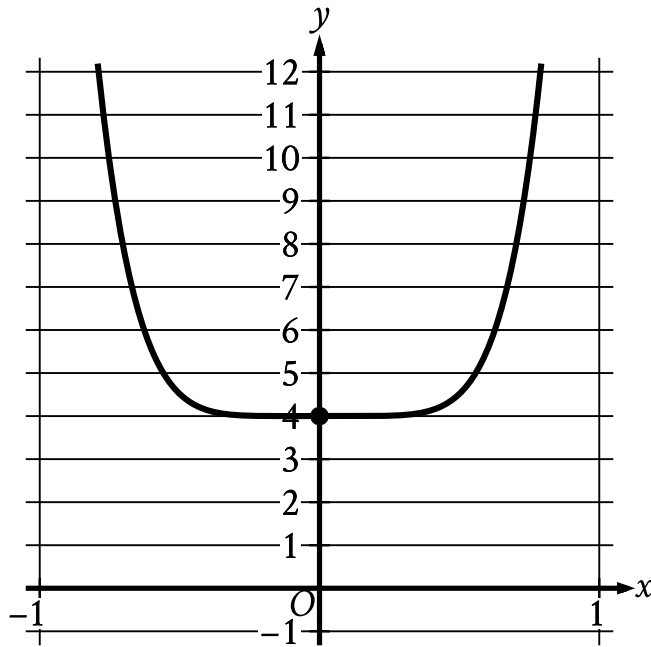
- The parabola opens upward.
- The vertex is at the point (4 comma 2).
- The parabola passes through the following points:
  - (0 comma 8)
  - (4 comma 2)
  - (8 comma 8)

The graph shows a marble's height above the ground  $y$ , in inches,  $x$  seconds after it started moving on an elevated track of a marble run. Which of the following is the best interpretation of the  $y$ -intercept of the graph?

- A. The marble's height was 0 inches above the ground 8 seconds after it started moving.
- B. The marble's height was 8 inches above the ground when it started moving.
- C. The marble's minimum height was 0 inches above the ground.



D. The marble's minimum height was 8 inches above the ground.



- In quadrant 2, the curve falls to its minimum at the point (0 comma 4).
- In quadrant 1, the curve then rises.

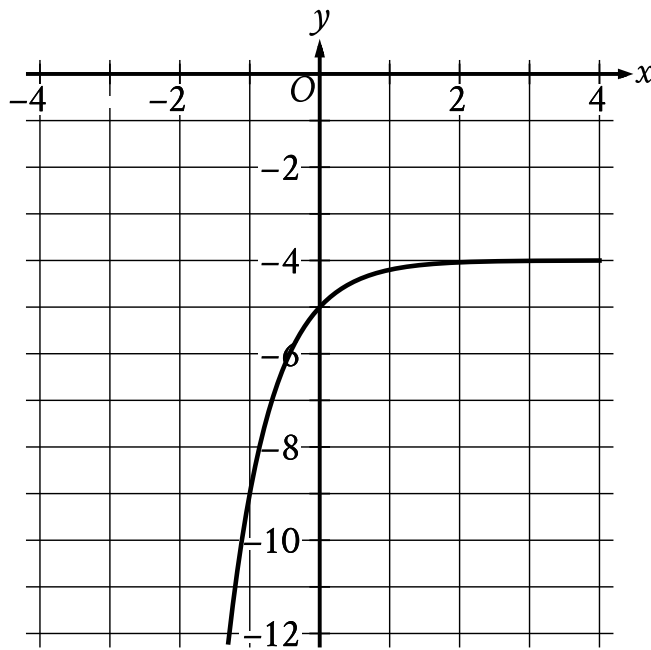
The graph of the polynomial function  $f$ , where  $y = f(x)$ , is shown. The  $y$ -intercept of the graph is 0,  $y$ . What is the value of  $y$ ?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
| 1                    | 1                    | 1                    | 1                    |
| 2                    | 2                    | 2                    | 2                    |
| 3                    | 3                    | 3                    | 3                    |
| 4                    | 4                    | 4                    | 4                    |
| 5                    | 5                    | 5                    | 5                    |
| 6                    | 6                    | 6                    | 6                    |

|   |   |   |   |
|---|---|---|---|
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Moving from left to right:
  - The curve passes from quadrant 3 to quadrant 4.
  - In quadrant 3, the curve trends up sharply to point (0 comma negative 5).
  - In quadrant 4, the curve trends up gradually.
- As  $x$  increases, the curve approaches the line  $y$  equals negative 4.
- The curve passes through the following points:
  - (negative 1 comma negative 9)
  - (0 comma negative 5)
  - (1 comma negative 4.2)

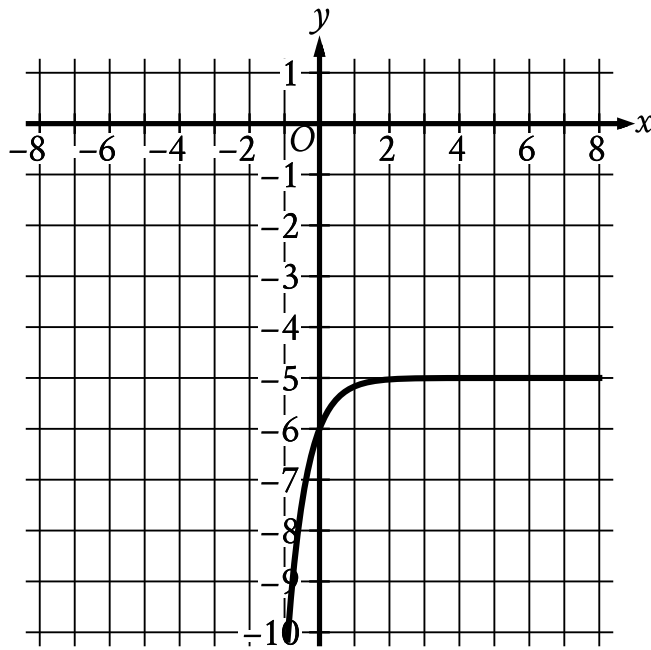
What is the  $y$ -intercept of the graph shown?

A. -1, -9

B. 0, -5

C. 0, -4

D. 0,0



- Moving from left to right:
  - The curve passes from quadrant 3 to quadrant 4.
  - In quadrant 3, the curve trends up sharply to point (0 comma negative 6).
  - In quadrant 4, the curve trends up gradually.
- As  $x$  increases, the curve approaches the line  $y$  equals negative 5.
- The curve passes through the following points:
  - (0 comma negative 6)
  - (1 comma negative 5 and one sixth)

What is the  $y$ -intercept of the graph shown?

- A. 0, -6
- B. -6, 0
- C. 0, 0

D. -5, - 5

Q9

ADVANCED MATH

The function  $f(x) = 200,000(1.21)^x$  gives a company's predicted annual revenue, in dollars,  $x$  years after the company started selling light bulbs online, where  $0 < x \leq 10$ . What is the best interpretation of the statement " $f(5)$  is approximately equal to 518,748" in this context?

- A. 5 years after the company started selling light bulbs online, its predicted annual revenue is approximately 518,748 dollars.
- B. 5 years after the company started selling light bulbs online, its predicted annual revenue will have increased by a total of approximately 518,748 dollars.
- C. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5 times the predicted annual revenue for the previous year.
- D. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5% greater than the predicted annual revenue for the previous year.

$$f(x) = 2(3^x)$$

For the function  $f$  defined above, what is the value of  $f(2)$ ?

- A. 9
- B. 12
- C. 18
- D. 36